Bilinear Parallel Fourier Transformer for Multimodal Remote Sensing Classification

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***Abstract*— Vision Transformers (ViTs) have shown promise in multimodal fusion image classification, yet face performance challenges in complex remote sensing scenarios. Single fusion frameworks often fail to fully utilize multimodal diversity, and the uneven distribution of image categories complicates the accurate construction of spatial structures by Transformers. Additionally, traditional cross-entropy tends to favor majority classes, neglecting minority classes, resulting in suboptimal predictions and reduced overall accuracy (OA). To solve these challenges, we propose a novel deep neural network, a bilinear parallel Fourier Transformer (BPFT). We propose a novel dual-fusion feature interaction (DFFI) module that utilizes two distinct types of fused features for learning, namely the spatial–spectral fusion feature and the global fusion feature. Besides, we introduce a dual-feature interaction (DFI) module to improve the utilization of fused feature information. To enable the Transformer to better establish spatial structural relationships, we employ the Fourier transform in place of the self-attention mechanism. To address the focus on minority class labels, we propose an exponential label smoothing cross-entropy loss function. This loss function comprises two components: exponential cross-entropy and label smoothing. The exponential cross-entropy component applies a strong penalty to misclassified samples, thereby increasing attention on minority class labels. To validate the efficacy of our approach, extensive experiments are conducted across two multimodal remote sensing datasets: Augsburg and Berlin, encompassing hyperspectral imaging (HSI) data and synthetic aperture radar (SAR) data. The results of these experiments affirm the superior performance of our proposed BPFT model compared to existing state-of-the-art models in multimodal remote sensing image classification tasks.**

***Index Terms*— Cross-modal retrieval, prior similarity, saliency learning.**

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I. Introduction

REMOTE sensing imagery is essential for capturing Earth's surface information via satellites, aircraft, or sensors. It plays a crucial role in land-cover classification [1], [2], environmental monitoring [3], resource management [4], urban planning [5], and disaster management [6]. Advancements in hyperspectral imaging (HSI) have enhanced remote sensing with high spectral resolution and rich data, yet challenges in distinguishing similar spectral responses in complex scenes, such as terrain variations, can be mitigated by integrating synthetic aperture radar (SAR) data. Consequently, combining HSI and SAR images shows promise for improving remote sensing analysis and applications [7].

Research highlights feature fusion as the key approach for multimodal integration in this field [8], [9]. The fusion stage can be categorized into three types: early, middle, and late fusion [10]. Studies by Hong et al. [11] and Wang et al. [12] suggest that early and late fusion methods may limit the interaction of information both within and across modalities. Due to its adaptability to deep learning, intermediate fusion has become the preferred method for multimodal fusion. Hong et al. [11] introduced an end-to-end unified deep learning model for remote sensing image classification, comprising two distinct architectures: ExNet for feature extraction and Fu-Net for data fusion. In subsequent work, Hang et al. [13] proposed an unsupervised coupled convolutional neural network (CNN) framework for hyperspectral and LiDAR data. In this framework, each modality serves as input to a separate CNN, with the coupled convolution mechanism facilitating mutual learning between the networks. This interaction aids in the effective fusion of the multimodal data. This article focuses on developing a novel feature fusion framework to enhance the model's feature extraction and fusion capabilities for HSI and SAR images.

Recently, Transformer networks have been introduced into the field of multimodal remote sensing data classification,

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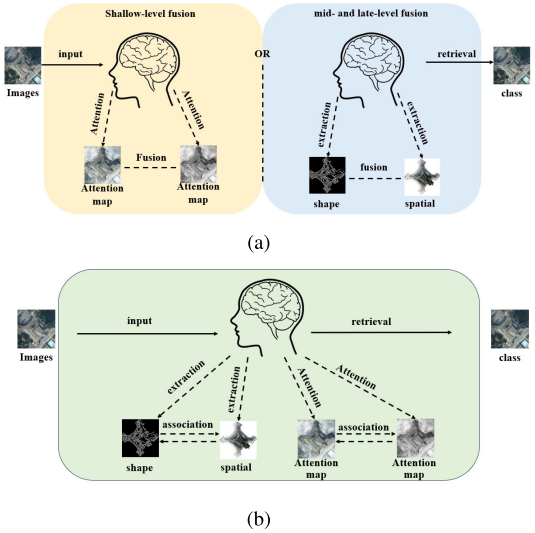


Fig. 1. Comparison of (a) triple-level fusion (early-level fusion, mid-level fusion, and late-fusion) and (b) our DFI module. Our module approach identifies the shape and spatial proportions of the primary target in the images to roughly determine the target's appearance. The regions associated with the primary target are then focused on. Ultimately, a comprehensive analysis of the collected multidirectional data enables more precise identification of surface cover and targets.

serving as a robust tool for multimodal feature extraction and multimodal heterogeneous feature fusion, owing to their distinctive and powerful ability to model long-range dependencies [14]. Deng et al. [15] introduced a robust joint framework amalgamating CNN and Transformer architectures to bolster the discriminative prowess of remote sensing image features. Lv et al. [16] devised the spatial channel feature preserving Vision Transformer (SCViT) model, which incorporates intricate geometric details from high-spatial-resolution (HSR) images alongside the distinct contributions of individual channels encapsulated within classification tokens. Sha and Li [17] presented the multi-instance Vision Transformer (MITformer), offering a solution to the challenge of Vision Transformer (ViT) overlooking crucial local features. Tang et al. [18] proposed the efficient multiscale Transformer and cross-level attention learning (EMTCAL) model, in which the multiscale Transformer (EMST) extracts rich knowledge across various scales in remote sensing scenes while capturing their intrinsic relationships. Yang et al. [19] developed a spatial frequency multiscale Transformer framework comprising branches dedicated to spatial and frequency domains, enabling comprehensive exploration of structural information at different scales. However, despite these innovations, many of the improvements to Transformers overlook the issue of uneven sample distribution, which can significantly affect model performance. In this article, greater emphasis is placed on enhancing the classification performance of remote sensing images in scenarios characterized by uneven sample distribution.

The main challenges in multimodal remote sensing image classification can be summarized in three key areas.

1) The two common feature fusion methods, concatenation-based fusion and alignment-based fusion [8], do not fully leverage the diversity of multimodal information. This approach, which relies on a single fusion feature branch, often leads to oversaturation.
2) In remote sensing datasets, the uneven distribution of samples across different categories creates difficulties in using Transformers for image classification. This imbalance complicates the construction of spatial structural information in complex scenarios, which negatively impacts overall performance.
3) Traditional cross-entropy loss tends to favor majority classes, neglecting minority classes. This imbalance in attention reduces classification accuracy. To improve performance, it is crucial to use a loss function that ensures balanced attention across all categories.

The current state-of-the-art methods, such as MFT [20], Cross-HL [21], ExViT [22], fusion-HCT [23], and SSMAE [24], use Transformers as their backbone to efficiently capture and learn the structural characteristics of the feature space. This helps optimize information extraction from modal features and cross-modal correlations. However, despite their strong performance in feature fusion, they are constrained by relying on a single fusion architecture, which often results in overly homogeneous feature information. Additionally, these methods fail to account for the impact of dataset sample distribution on the Transformer's feature space construction. To address these limitations, our proposed model tackles the problem of single-type fusion features and mitigates the negative effects of uneven sample distribution on the Transformer architecture.

Effectively addressing multimodal classification problems frequently depends on the ability to analyze and process a diverse array of heterogeneous images [25]. The remote sensing images vary in spatial, spectral, angular, as well as temporal resolutions, necessitating sophisticated techniques to manage their differences and extract meaningful information. As shown in Fig. 1(a), in the traditional way of processing heterogeneous images, the spatial and shape information of multiple modalities is often directly combined, or the major components are extracted first and the minor components are ignored before fusion analysis. Fig. 1(b) illustrates the underlying concept of our proposed strategy. Upon acquiring various types of images, we initially identify the shape and spatial proportion of the primary target in these images, thereby estimating the target's appearance information. Subsequently, we focus on the regions related to the primary target to enhance recognition accuracy. In summary, our main idea is to maximize the extraction of unique feature information from diverse types of remote sensing data and integrate them comprehensively to enhance target recognition accuracy.

To address the aforementioned challenges, we propose a bilinear parallel Fourier Transformer (BPFT) network to explore and integrate multimodal semantic information. Our method aims to acquire diverse fusion features, yielding richer multimodal semantic characteristics. In the proposed framework, our spatial–spectral fusion feature extractor module and global fusion feature extractor (GFFE) module obtain two complementary fusion features, which undergo Fourier transform blocks for deep feature learning, thereby extracting features from images at multiple scales. Subsequently,

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we employ a dual-feature interaction (DFI) module that utilizes sampling operations for feature interaction, achieving a complementary effect in feature representation. The architecture of our BPFT network for HSI and SAR data classification is illustrated in Fig. 2. In summary, our work contributes significantly as follows.

1) We introduce a novel dual-fusion feature interaction (DFFI) framework capable of simultaneously learning two distinct types of features. This represents a pioneering approach to feature fusion in multimodal remote sensing image classification.
2) We propose a novel Fourier Transformer encoder that integrates the Fourier transform into the attention mechanism of the Transformer. This transforms images from the time domain to the frequency domain for feature learning, enhancing the model's generalization ability to the data.
3) We present a novel DFI module designed to facilitate feature fusion by leveraging complementary information during the simultaneous learning of two fusion features.
4) We propose a novel exponential label smoothing cross-entropy loss function based on traditional cross-entropy. This new loss function adjusts attention toward minority class labels by incorporating exponential functions and label smoothing.
5) We perform comprehensive experiments on two publicly available remote sensing datasets: Augsburg and Berlin. These experiments demonstrate the efficacy of the proposed method.

The rest of this article is listed as follows. Section II introduces the related work. Section III introduces the details of the proposed BPFT network. Section IV lists quantities of comparative experiments, ablation experiments, and parameter experiments for BPFT. Section V summarizes with some concluding remarks on BPFT.

II. Related Work

A. Transformer-Based Methods

Transformers stand out as cutting-edge backbone networks due to their utilization of self-attention techniques, highly effective for processing and analyzing sequential or time-series data, unlike CNNs and recurrent neural networks (RNNs). Inspired by the transformative success of Transformers in computer vision, numerous novel Transformer models have emerged in recent years. Hong et al. [26] introduced the spectral former, a recent addition capable of learning spectral representation from adjacent borehole bands and constructing cross-layer Transformer encoder modules. They also proposed the spectral–spatial feature tokenization (SSFT) Transformer model [27], leveraging the Transformer's prowess in extracting local spatial semantic information and establishing connections between adjacent sequences. However, a notable limitation of SSFT lies in its parameter count. While SSFT demonstrates impressive performance on HSI data, it falls short in integrating other multimodal data sources during classification tasks. Ding et al. [14] presented the global–local Transformer network (GLT-Net) to fully exploit the Transformer architecture's potential in modeling long-range dependencies and convolutional operators. These operators play a crucial role in characterizing locally relevant features, particularly in the joint classification of HSI and SAR data. To enhance joint classification performance and fuse heterogeneous information sources, Zhao et al. [23] proposed an innovative dual-branch network comprising a Transformer network and hierarchical CNNs. For decades, the Fourier transform has been a pivotal tool in digital image processing. Yu et al. [28] developed the cross-context capsule ViT (CapViT) to utilize global feature interactions for land-cover classification with MS-LiDAR data. The local information interaction Transformer (LIIT) was introduced to combine HSI and LiDAR data, capturing relationships and fusing features without redundancy [29]. The multiattentive hierarchical fusion network (MAHiDFNet) was also created for feature-level fusion and classification of HSIs with LiDAR data [30]. The morphFormer model [31] improved the connection between HSI tokens and class tokens using spectral and spatial morphological convolution along with a self-attention mechanism for land-cover classification.

The methods mentioned earlier have modified the Transformer architecture and integrated it with convolutional networks to improve performance in the multimodal domain. While the self-attention mechanism of the Transformer helps establish internal relationships between modalities, it requires a large volume of samples for effective fitting due to the significant number of parameters involved. This requirement is particularly challenging given the uneven distribution of sample categories in remote sensing images, which negatively impacts the Transformer's performance. To tackle this issue, the Fourier transform [32] has been employed to reduce the number of model parameters, thereby decreasing the dependency on extensive data samples.

B. Conventional Modal Fusion Methods

Lately, there has been a surge of research [33] focusing on effectively extracting and comprehending the fundamental interrelations among multimodal features, to integrate them to enhance the precision of remote sensing image classification. Du and Zare [34] proposed a novel framework called multiple instance multisolution fusion (MIMRF), which is capable of fusing multisolution and multimodal sensor outputs while learning from automatically generated, accurately labeled data. Liu et al. [35] designed a CNN extractor to combine visual and multimodal information, feeding the features into a graph convolutional network (GCN) extractor to obtain GCN-based features that capture correlations among multimodal cloud samples. Additionally, Du et al. [36] introduced an unsupervised feature extraction network for HSIs and LiDAR, in which a multimodal graph is constructed for feature fusion. Mei et al. [37] presented a sparse representation-based framework for small-sample-size remote sensing scene classification, employing deep feature fusion to integrate multilevel features, particularly beneficial when dealing with limited training samples. Li et al. [38] introduced a multimodal fusion architecture search network to address

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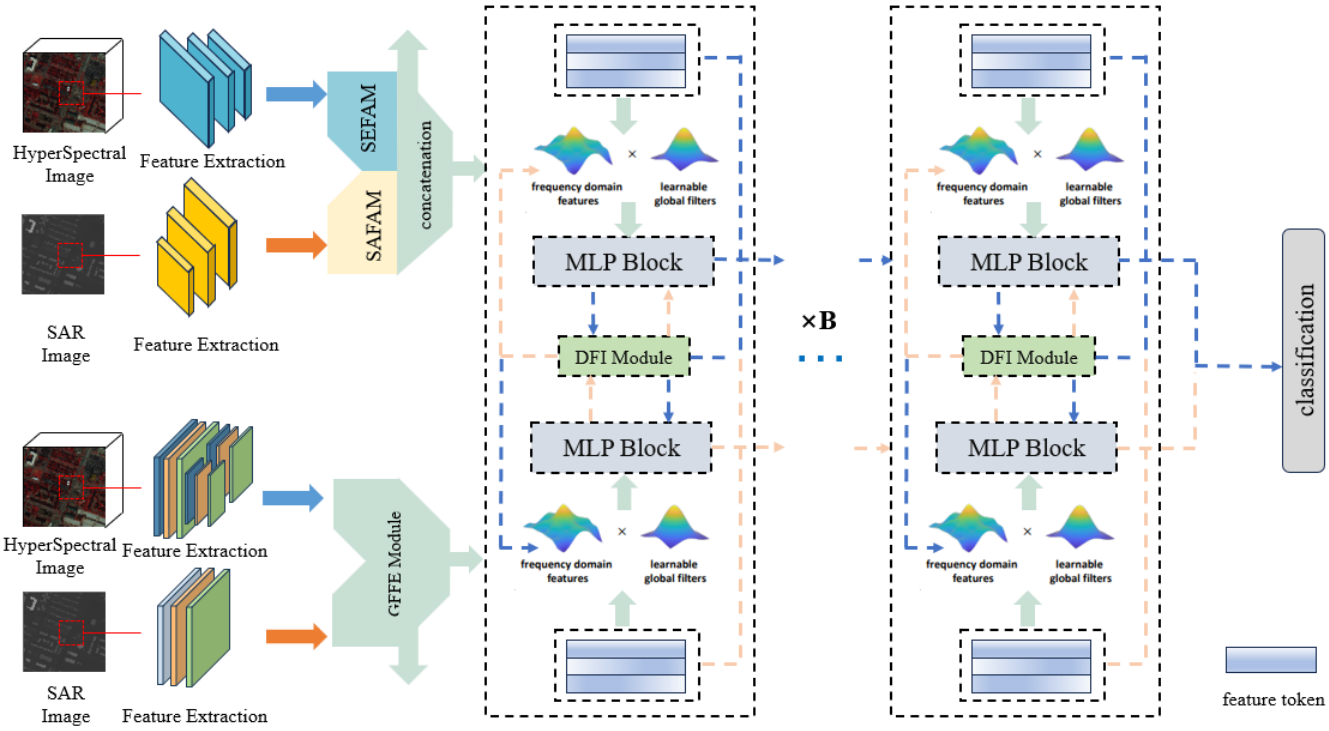


Fig. 2. Overview of the BPFT framework for HSI and SAR data classification. The model is divided into two stages: feature extraction and interaction. In the feature extraction stage, spatial–spectral fusion features and global fusion features are obtained using the spatial–spectral fusion feature extractor module and the GFFE module. The Fourier transform is then applied to analyze the spatial structures of these features. In the feature interaction stage, a DFI module enables information interaction between the two fusion features, enhancing the model's learning efficiency.

the challenge of identifying optimal fusion architectures for multimodal land-cover classification. Wu et al. [8] proposed a new deep learning-based framework for multimodal remote sensing data classification, utilizing CNNs as the backbone and incorporating an advanced cross-channel reconstruction module. He et al. [47] proposed a foundation model adaptation (FMA) framework for multimodal remote sensing data classification that does not require fine-tuning the parameters. Instead, two learned modules are added to the foundation model to extract multimodal-specific representations.

Despite the advancements in feature extraction and other methods that integrate more comprehensive information to enhance model performance, making dynamic adjustments based on varying information across modalities in complex and changing scenarios remains a significant challenge. Consequently, a universal framework is essential to fully utilize the effective information in each modality.

III. Methodology

In this section, we will describe the details of the BPFT. The overall framework consists of two fusion feature branches, namely the spatial–spectral fusion feature branch and the global fusion feature branch. Each branch operates in two stages: shallow feature extraction and deep feature interaction. The spectral cross-modality enhancement module (SEEM) and the spatial cross-modality enhancement module (SAEM), respectively, integrate spectral and spatial features between HSIs and SAR. The GFFE module handles the extraction of global fusion features from both HSIs and SAR. The DFFI module is used to process deep features. To achieve feature information interaction between two branches, we use the DFI module to fuse the features of the two branches.

Let $\mathcal{X}_{\text{HSI}} \in \mathbb{R}^{H\times W\times C_H}$ and $\mathcal{X}_{\text{sar}} \in \mathbb{R}^{H\times W\times C_S}$ denote the raw HSI and SAR data, respectively, which C_H and C_S are the channel dimensions. In the data preprocessing stage, we segment the HSI and SAR images into patches of the same length and width, $x_h^{i,j} \in \mathbb{R}^{p\times p\times C_H}$ and $x_s^{i,j} \in \mathbb{R}^{p\times p\times C_S}$, which denotes that the patch size is p and the patch centers at pixel (i, j).

A. Spatial–Spectral Fusion Feature Extractor

To effectively fuse HSI data, which often comprises numerous channels, with SAR images that have significantly fewer channels, feature alignment procedures are necessary. Incorrect initialization values during feature alignment can lead to suboptimal outcomes. A high initialization value may introduce excessive redundancy in SAR image features, while a low one risks losing valuable information from the HSI features. Thus, precise calibration of initialization values is crucial to ensure the successful fusion of features across these diverse imaging modalities.

To solve the above problems, we adopt a three-layer alignment module to achieve feature alignment. The first layer uses ordinary 2-D convolution operations, while the second and third layers use residual blocks to reduce information loss during the feature alignment process. After feature alignment, we obtain a feature vector with a channel number of $d_m = 32$.

1) SAEM: Because of the lack of sufficient spatial information in HSIs, we use SAEM [39] to introduce spatial

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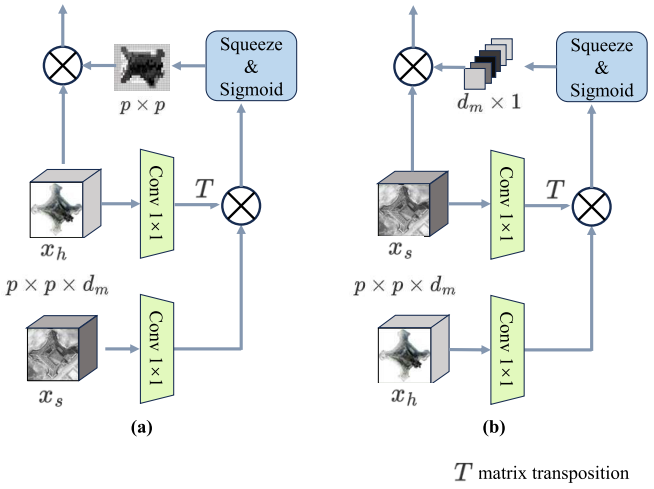


Fig. 3. Architecture of (a) SAEM and (b) SEEM.

information from SAR images into the HSI. The definition of SAEM is as follows:

$$z_i^{(\mathrm{SA})} = \mathrm{Squeeze}\left(f\left(x_h^i, x_s^0\right), \ldots, f\left(x_h^i, x_s^j\right), \ldots, f\left(x_h^i, x_s^{t_n-1}\right)\right) \tag{1}$$

$$f(x_h, x_s) = \mathcal{T}^T(x_h)\mathcal{T}(x_s) \tag{2}$$

$$x_h{}' = z_i^{(SA)} \otimes x_h \tag{3}$$

where $x_s \in \mathbb{R}^{p\times p\times d_m}$ is the input data from HSIs and $x_s \in \mathbb{R}^{p\times p\times d_m}$ is the input data from SAR. $x_s^i \in \mathbb{R}^{d_m\times 1}$ and $x_s^j \in \mathbb{R}^{d_n\times 1}$ denote the vectors at position i of x and position j of y. $t_n = p^2$ means the size of y. The function of the squeeze is to combine a series of spatially enhanced data $z_i^{(SA)}$ into a scalar. $\mathcal{T}$ consists of a dot product and a learnable 1×1 convolutional layer. x_h' means the data after spatial enhancement processing.

2) SEEM: Similarly, because SAR data has a limited number of channels and lacks sufficient spectral information, we use the SEEM [39] to introduce spectral features from HSI data into SAR data. The SEEM is defined as

$$z_i^{(\mathrm{SE})} = \mathrm{Squeeze}\left(f\left(x_s^i, x_h^0\right), \ldots, f\left(x_s^i, x_h^j\right), \ldots, f\left(x_s^i, x_h^{t_n-1}\right)\right) \tag{4}$$

$$x_s' = z_i^{(\mathrm{SE})} \otimes x_s \tag{5}$$

where $x_s^i, x_h^j \in \mathbb{R}^{p^2\times 1}$ denotes the flattened vectors from SAR and HSI features and i means the ith channel in data y and j means the jth channel in data x. The output $z_i^{(\mathrm{SE})}$ shape is $d_m \times 1$, which means the connection between every channel of SAR and all channels of HSIs. The structure of the above SAEM and SEEM modules is shown in Fig. 3.

B. Global Fusion Feature Extractor

Although the relationship between HSIs and SAR is established in space and spectrum, it is not enough to only obtain the spectral–spatial enhancement features between HSIs and SAR, and the relationship between modalities also needs to be considered.

HSIs not only contain spatial information (height and width) but also spectral information. Two-dimensional convolution extracts features from each channel independently, which leads to poor correlation between different channels. On the other hand, 3-D convolution can capture both spatial and spectral information and share information between channels, preserving more correlation between channels.

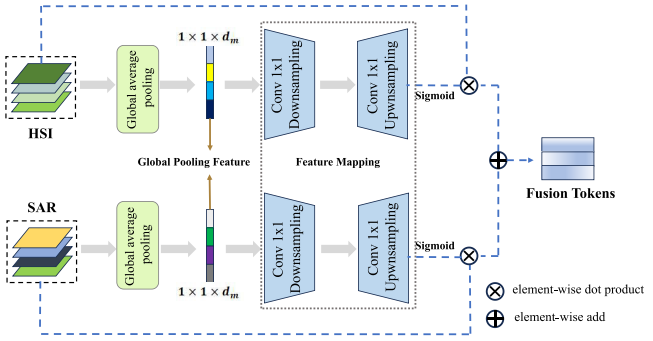


Fig. 4. Detail of the GFFE module.

To extract richer features, reduce information loss, and increase model robustness, 3-D convolution is used for processing the original HSI. First, because the shape of input data is (p, p, C_H), we need to unsqueeze it into shape $(1, p, p, C_H)$ and pass it into first Conv3D layer, which kernel size is $(3 \times 3 \times 9)$, and to keep length and width unchanged we let the padding is $(1 \times 1 \times 0)$. After that, we need to align the dimension, so we utilize HetConv2d [40] to finish feature dimension reduction. The HetConv2d utilizes two Conv2D operations in parallel, one of which performs a groupwise conv2D (kernel size $= 3$, group $= 4$, and padding $= 1$). The results of two Conv2D are added in an elementwise ($\oplus$) fashion.

All of these layers are followed by batch normalization [41] and rectified linear unit (ReLU). For SAR, we just utilize a Conv2D layer to align its dimension with HSIs.

After feature alignment, we obtain a global pooling feature for both branches through a global average pooling layer. Next, we apply a feature mapping operation consisting of two Conv2D layers with convolutional kernel size 1 to the globally pooled features, which is conducive to increasing the receptive field of the feature map, allowing the network to learn more extensive features during later stages while preserving advanced features and details, ultimately enhancing the model's performance. We use the Sigmoid function to get enhanced features and finally weight and add two branch features element-wise to get a global fusion feature. The procedure is shown in Fig. 4.

C. DFFI Module

After a series of operations, we acquire two main fusion feature branches, which are the spatial–spectral fusion feature and the global fusion feature. We propose two modules for deep feature learning and interaction, as follows.

1) Fourier Transformer Encoder: Incorporating complementary information can substantially improve a model's capacity to learn discriminative features. Moreover, the generalized architecture of the converter facilitates the integration and processing of heterogeneous multimodal data, including

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HSIs with detailed spectral information, RGB images with HSR, and SAR data with elevation attributes [42]. However, integrating this into a conventional Transformer architecture results in an exponential increase in the number of parameters [20]. Additionally, the high dimensionality of spectral bands in the input exacerbates the complexity of the linear projection, increasing the risk of overfitting.

To address the aforementioned challenges, we propose a Fourier Transformer encoder as a replacement for the encoder in the conventional Transformer architecture. This approach eliminates the self-attention mechanism and leverages the discrete Fourier transform (DFT) to extract and analyze features. Compared to the self-attention mechanism, DFT significantly reduces the computational overhead of the Transformer. Furthermore, the integration of DFT with multilayer perceptron (MLP) layers enhances the Transformer's ability to effectively capture both contextual and spectral information across patches.

We replace the self-attention module in the traditional ViT with 2-D DFT [32] to obtain the attention weight in different frequency domains. Given $x \in \mathbb{R}^{p^2 \times C}$, which is x of size $(p \times p \times d_n)$, it is flattened into feature patch tokens with a shape of $(p^2 \times C)$. To maintain the correlation between spatial locations, trainable position embeddings are added to the patch embeddings in an element-wise manner. This process is followed by a dropout operation to reduce the risk of overfitting.

Given the 2-D signal $X[m, n]$, $0 \le m \le M-1, 0 \le n \le N-1, M = p, N = p$, the 2-D DFT is defined by

$$X[u, v] = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} x[m, n] e^{-j2\pi\left(\frac{um}{M} + \frac{vn}{N}\right)}. \tag{6}$$

While the Fourier transform effectively analyzes the spectral characteristics of signals, its computational complexity is O (n^2), where n is the length of the signal. Consequently, for large signals, the computation time increases significantly. Fast Fourier transform (FFT) [43] is an optimized algorithm that accelerates the computation of the Fourier transform. By leveraging the symmetry and periodicity of signals, it reduces the computational complexity of the Fourier transform to o($n$log$n$), greatly enhancing computational efficiency

$$X = \mathcal{F}(x) \in \mathbb{C}^{p \times p \times d_n} \tag{7}$$

where $\mathcal{F}(x)$ denotes the 2-D FFT and X is a complex tensor. We can get an attention matrix by multiplying a learnable filter $K \in \mathbb{C}^{p \times p \times d_n}$ to X

$$\bar{X} = X \odot K, K \in \mathbb{C}^{p \times \widehat{p} \times d_n}, \widehat{p} = \lceil p \rceil = p//2 + 1 \tag{8}$$

where $\odot$ means elementwise multiplication. It is worth noting that after using the FFT, $\bar{X}$ is conjugate symmetric, that is, $\bar{X}[p-u, p-v, :] = \bar{X}^*[p, p, :]$. Therefore, we only need to extract half of the x values and retain all the frequency information. We can implement learnable filter K as $\mathbb{C}^{p \times p \times d_n}$. Finally, we need to adopt the inverse FFT to transform X to its original spatial domain

$$\begin{aligned} x &= \mathcal{F}^{-1}\left(\bar{X}\right) \\ x_k &= \text{FourierAttention}(\text{LN}(x_{k-1})), \quad 1 \le k \le L. \end{aligned} \tag{9}$$

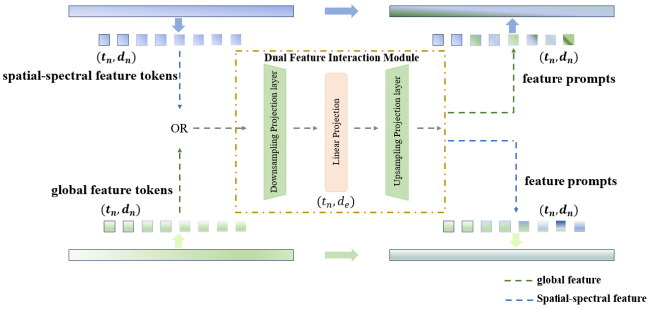


Fig. 5. Detail of the DFI module. It consists of three linear layers. t_n denotes the feature tokens, the input dimension d_n is reduced to d_e and passed through a linear layer, and then restore the dimension to the original dimension d_n by upsampling the linear layer, and is added to another modality feature.

After obtaining the data extracted from the spectrum features, we need to reshape it into the shape $(p^2 \times d_n)$ so that it can be processed by the following MLP layers. The data x is processed by a residual shortcut, where k denotes the kth DFFI block.

The utilization of the Fourier transform instead of the self-attention module in the Transformer architecture reduces computational complexity [32]. This reduction in computational overhead not only facilitates scalability to deeper networks but also signifies a methodological advantage over traditional ViTs. Importantly, this approach enhances the preservation of the structural integrity of images, highlighting its importance in image processing tasks.

2) DFI Module: Multimodal feature interaction plays an important role in multimodal learning. Traditional methods, such as simple addition or concatenation, often treat one mode as primary and the other as auxiliary. This results in an incomplete fusion of auxiliary and primary features during learning. The purpose of the DFI module is to make the two fusion features the primary features at the same time, enabling full interaction between them throughout the learning process.

The DFI module is shown in Fig. 5. The DFI module is composed of three linear layers. The first linear layer reduces the feature dimension to d_m and the third linear layer increases the feature dimension back to the original dimension. Since the features of X and Y contain different information, it is essential to extract complementary features through downsampling and upsampling during feature interaction. The DFI module is embedded in both the Fourier attention stage and the MLP stage.

Here, we denote x_i^{SASE} as the ith layer spatial–spectral feature and x_i^{GF} as the ith layer global fusion feature. The procedure that the ith layer of spatial–spectral feature integrates auxiliary from the global feature is shown as

$$\begin{aligned} x_i^{\text{SASE}'} &= x_i^{\text{SASE}} + \mathcal{F}^{\text{FAtt}}\left(LN\left(x_i^{\text{SASE}}\right)\right) + B_i^{\text{GF}} \\ B_i^{\text{GF}} &= \mathcal{F}^{\text{DFI}}\left(x_i^{\text{GF}}\right), i = 1, 2, \ldots, N \end{aligned} \tag{10}$$

where $\mathcal{F}^{\text{FAtt}}$ is the Fourier attention block and $\mathcal{F}^{\text{DFI}}$ is the DFI block. LN$(\cdot)$ is the LayerNorm operation. B_i^{GF} is the feature extracted from x_i^{GF} by the DFI block.

In the next stage, the $x_i^{\text{SASE}'}$ is fed into MLP layers and added with $B_i^{\text{GF}'}$ and x processed by LayerNorm to get the

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output of x_{i+1}^{SASE}

$$x_{i+1}^{\mathrm{SASE}} = x_i^{\mathrm{SASE}'} + \mathcal{MLP}\left(LN\left(x_i^{\mathrm{SASE}'}\right)\right) + B_i^{\mathrm{GF}'}$$
$$B_i^{\mathrm{GF}'} = \mathcal{F}^{\mathrm{DFI}}\left(x_i^{\mathrm{GF}'}\right). \tag{11}$$

Following multiple iterations through Fourier Transformer layers, the resulting outputs are aggregated to derive fused features. These fused features undergo preprocessing for classification, including passing through a global average pooling layer followed by a linear layer

$$\mathcal{Z}_b = x_b^{\mathrm{SASE}} \oplus x_b^{\mathrm{GF}}$$
$$\mathcal{Z}_{\mathrm{OUT}} = \mathcal{L}(\mathrm{Unsqueeze}(\mathrm{GAP}(\mathcal{Z}_b))) \tag{12}$$

where b means the number of DFFI blocks, GAP denotes the global average pool, and $\mathcal{L}$ is the linear layer.

D. Exponential Label Smoothing Loss Function

Although the traditional cross-entropy has the advantages of good classification performance and wide applicability in the field of image classification, it struggles with more complex scenarios, particularly when dealing with uneven sample distributions. In remote sensing images, the presence of complex geological environments often leads to imbalanced classification samples, as well as inevitable noise in the images. Under these challenging conditions, traditional cross-entropy has difficulty achieving optimal results

$$\mathrm{ExpLoss} = e^{-\left(\sum_{i=1}^{N} y_i^{\mathrm{GT}} \cdot \log(y_i)\right)/\alpha}. \tag{13}$$

The exponential loss function is modified in the traditional cross-entropy, as shown in (13), where N is the number of classes, y_i is the predicted value, y_i^{GT} is the ground-truth label, and α indicates the penalty applied to the model. This modification makes the exponential loss function more sensitive to misclassified samples. The farther the predicted value deviates from the real label, the greater the penalty imposed by the loss function. This forces the model to focus more on complex samples and improves the generalization ability of the model.

The conventional cross-entropy method often faces challenges in effectively classifying intricate scenes characterized by noise, erroneous label interference, and other complexities. Label smoothing cross-entropy (14) can deal with this disadvantage. Instead of being strictly 0 or 1, the labels become continuous values between 0 and 1. This adjustment reduces overfitting to the training data and enhances the model's robustness to noise

$$\mathrm{SmoothLoss} = \sum_{i=1}^{N} y_i^{\mathrm{GT}} \cdot \log \frac{e^{y_i}}{\sum_j e^{y_j}} (1-\beta) + \beta/K \tag{14}$$

where β is used to control the smoothness of labels when β is close to 1, it indicates strong smoothing of the labels. When β is close to 0, it means that no smoothing is performed, (β/K) represents the average value of the smoothing parameters in the loss function, ensuring that the expected value of the loss function remains constant across the training set. This prevents deviations caused by changes in the number of samples

$$\mathrm{ESLoss} = e^{-\left(\sum_{i=1}^{N} y_i^{\mathrm{GT}} \cdot \log \frac{e^{y_i}}{\sum_j e^{y_j}}/\alpha\right)} (1-\beta) + \beta/K. \tag{15}$$

According to the characteristics of the above two loss functions, we propose a new loss function ESLoss, as shown in formula (15). The formula can well deal with the shortcomings of uneven sample classification, complex scenes, and noise in remote sensing images

$$\mathcal{LOSS} = a \times (\mathrm{ESLoss}(\mathcal{Z}_{\mathrm{OUT}})) + b \times \left(\mathrm{ESLoss}\left(x_0^{(\mathrm{SASE})}\right)\right) + c \times \left(\mathrm{ESLoss}\left(x_0^{\mathrm{GF}}\right)\right).$$

We adopt a strategy in model training that involves incorporating multibranch loss functions by introducing separate branches for loss function computation after the SFFE and GFFE modules. This approach offers several advantages. First, it enables the model to capture diverse levels of information, allowing for a more comprehensive understanding of the data. Second, it helps reduce overfitting by encouraging the model to learn generalized features, thereby reducing the risk of excessively fitting to the training data. This multibranch loss function method enhances both the model's ability to learn nuanced information and its generalization capabilities.

TABLE I
NUMBER OF TRAINING AND TEST SAMPLES FOR THE AUGSBURG DATASET

No.	Color	Name	Train	Test
1		Forest	146	13361
2		Residential area	264	30065
3		Industrial area	21	3830
4		Low Plants	248	26609
5		Allotment	52	523
6		Commercial area	7	1638
7		Water	23	1507
Total			761	77533

IV. EXPERIMENT

The server of the testing environment for our study is equipped with a 96-core CPU and 50 GB of memory. Additionally, it features an NVIDIA Tesla A100 graphics card with 40 GB of VRAM. The source code for the proposed BPFT model was implemented using PyTorch version 2.1.2, Python 3.13, and CUDA version 12.3.

A. Dataset Description

In this section, we will evaluate our proposed model BPFT in the following two remote sensing datasets: Augsburg and Berlin. The Augsburg and Berlin datasets are used for HSI and SAR data classification.

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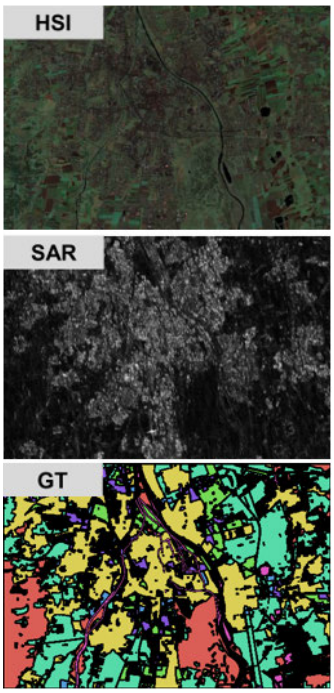


Fig. 6. Visualization of the Augsburg dataset. The HSI is the true-color image using bands 40, 20, and 10, respectively. The SAR is a grayscale image. The GT is the ground truth.

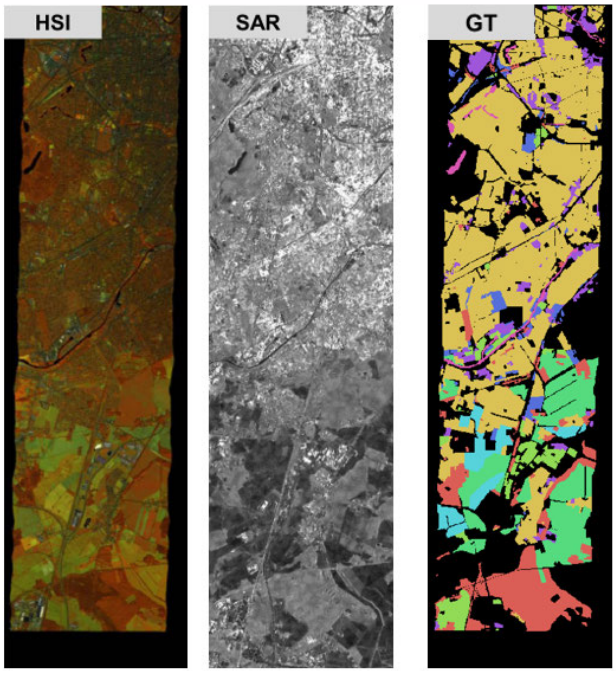


Fig. 7. Visualization of the Berlin dataset. The HSI is the true-color image using bands 40, 20, and 10, respectively. The SAR is a grayscale image. The GT is the ground truth.

1) Augsburg Dataset: This dataset consists of HSI and SAR data from Augsburg, Germany. The HSI is acquired using the airborne imaging spectrometer system HySpex. The SAR data is acquired by the Sentinel-1 sensor. All images are with a ground sample distance (GSD) of 30 m. The size of HSI and SAR images is 332 × 485 pixels [44]. The HSI has 80 bands ranging from 0.4 to 2.5 μm. The SAR data has four features (VV intensity, VH intensity, real part, and imaginary part of the off-diagonal element of the polarimetric SAR covariance matrix).

TABLE II
NUMBER OF TRAINING AND TEST SAMPLES FOR THE BERLIN DATASET

No.	Color	Name	Train	Test
1		Forest	443	54511
2		Residential area	423	268219
3		Industrial area	499	19067
4		Low Plants	376	58906
5		Soil	331	17095
6		Allotment	280	13025
7		Commercial area	298	24526
8		Water	170	6502
Total			2820	461851

2) Berlin Dataset: This dataset covers the urban and rural areas of Berlin. The HSI data is the data of simulated environmental mapping and analysis program (enmap) based on HyMap HS data. SAR data is a sentinel-1 dual-Pol Single Look Complex (SLC) product obtained from the European Space Agency (ESA). The HSI has 797 × 220 pixels and 244 spectral bands in the wavelength range of 400–2500 nm [45]. The size of the SAR image is 1723 × 476 pixels. The nearest neighbor interpolation method is used to match the spatial resolution of the two images.

Figs. 6 and 7 display the HSI and SAR with ground truth. We use overall accuracy (OA), average accuracy (AA), and statistical Kappa as the evaluation metrics. Specifically, OA is the proportion of accurately classified samples through the dataset, AA denotes the mean accuracy across all classes, and Kappa is a statistical measure of agreement between predicted and true labels. Tables I and II enumerate the number of samples for both training and testing on both Augsburg and Berlin datasets.

B. Experimental Setup

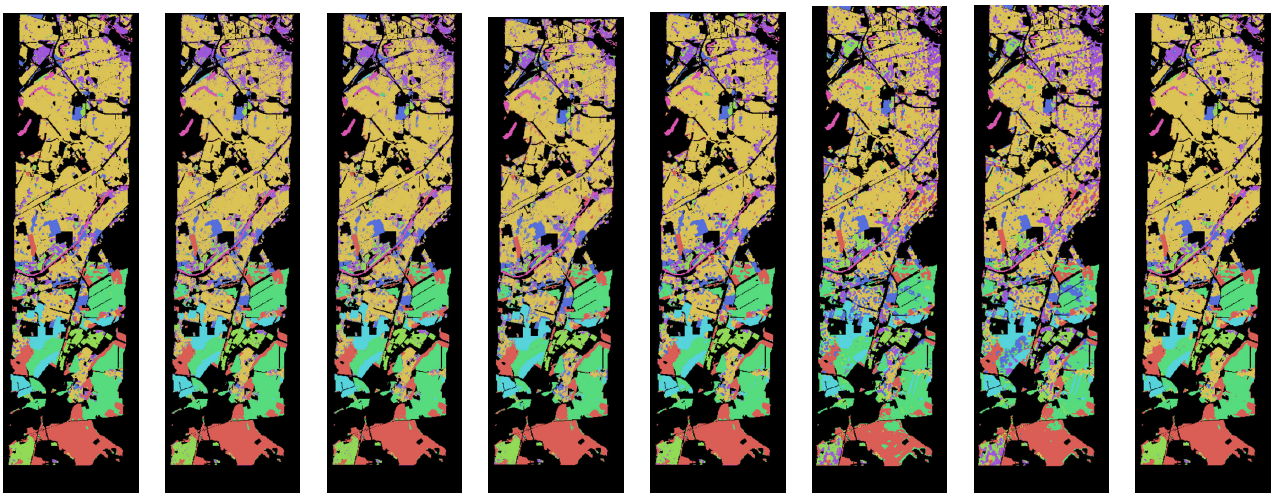
To evaluate the efficacy of the proposed BPFT model, we conduct a comprehensive set of experiments, contrasting it against both traditional and cutting-edge methodologies. Our comparative analysis encompasses established classifiers like SSMAE (with divisions for pretraining and nonpretraining), crossHL, fusion HCT, ExViT, MFT, and S^2ENet. Our experimental setup utilizes a combination of HSI data and SAR data.

C. Parameter Setting

This study focuses on investigating the impact of parameter configurations on the performance of remote sensing image classification tasks. Specifically, we conduct fine-tuning of two categories of parameters.

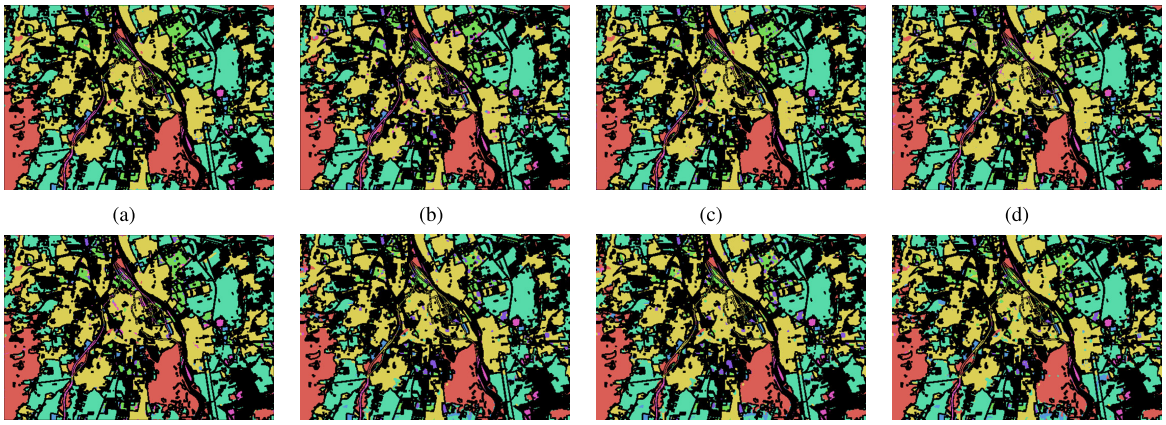
1) Network structure parameters, encompassing adjustments to the GFFE module, the sampling structure within the DFI module, and the number of DFFI blocks.
2) Loss function parameters, including tuning the weights (α and β) within the loss function and optimizing the weight ratios in the multibranch loss function.

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(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 8. Classification results of different methods for the Berlin dataset. (a) S^2ENet. (b) MFT. (c) CrossHL. (d) ExViT. (e) HCTNet. (f) SSMAE*. (g) SSMAE. (h) BPFT.



(a) (b) (c) (d)

(e) (f) (g) (h)

Fig. 9. Classification results of different methods for the Augsburg dataset. (a) S^2ENet. (b) MFT. (c) CrossHL. (d) ExViT. (e) HCTNet. (f) SSMAE*. (g) SSMAE. (h) BPFT.

This professional approach enables a detailed exploration of how these parameter settings affect the accuracy and efficacy of classification models in the context of remote sensing image analysis.

1) Training Parameter Configuration: The training and testing of the models under consideration utilize a batch size of 64 and 600, respectively. During training, patches of size $9 \times 9 \times C_H$ are extracted from the HSI, while patches of size $9 \times 9 \times C_S$ are extracted from other multimodal data sources. All models are trained using the Adam optimizer [46] with a learning rate of $1e^{-3}$. Specifically, for SSMAE, we adjust the learning rate to $1e^{-4}$ and apply a weight decay of $5e^{-2}$.

2) Block Number Comparison: We conduct experiments testing varying numbers of Fourier Transformer encoder block B, ranging from 4 to 8, to assess their impact on classification performance using the Augsburg and Berlin datasets. As illustrated in Fig. 10(a), the graph demonstrates the relationship between B and classification performance. Deriving from the results, increasing the number of DFFI modules enhances the model's capacity to capture a wider range of frequency features, thereby bolstering its generalization performance to a certain degree. However, this augmentation introduces challenges such as exponential parameter growth and susceptibility to overfitting, necessitating a careful balance between

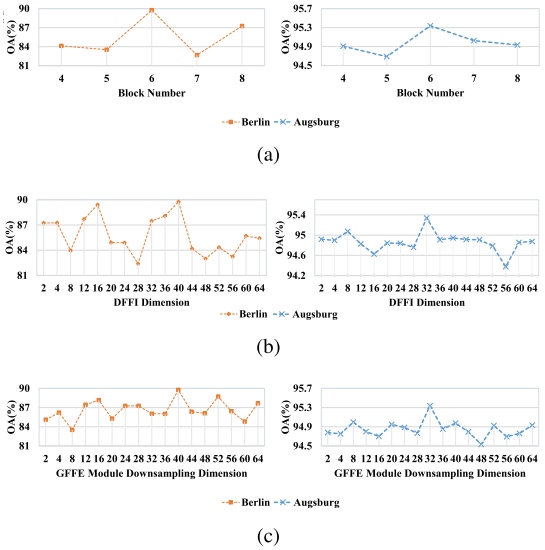
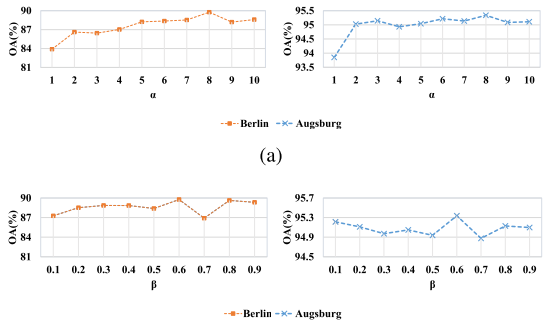


Fig. 10. OA (%) obtained by the different parameters of the BPFT model on the Augsburg dataset. (a) Block number. (b) DFI module downsample dimension. (c) GFFE module downsample dimension.

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(a)

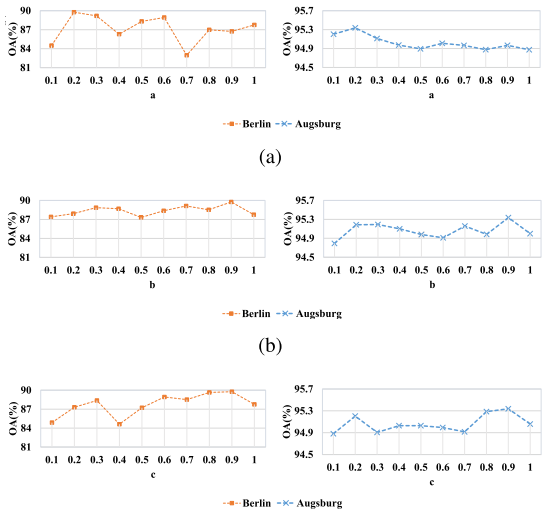
(b)

Fig. 11. OA (%) obtained by the different parameters of the ESLoss function on the Augsburg dataset. (a) Penalty weight. (b) Label smoothness.

model complexity and robustness. We can observe that $B = 6$ yields optimal results for the Augsburg and Berlin datasets.

3) Subsampling Dimension: The efficacy of feature extraction within modules is influenced by varying subsampling dimensions. In our study, we employ different subsampling dimensions within the GFFE module and the DFI module to evaluate their impact on model performance using the Augsburg and Berlin datasets. Specifically, we vary the subsampling dimension from 2 to 64. The downsampling dimension plays a pivotal role in feature extraction by enabling the module to focus on essential information while filtering out superfluous details. In the context of the GFFE module, downsampling facilitates the extraction of saliency features crucial for global representations, thereby enhancing the model's learning efficiency. Similarly, within the DFI module, downsampling facilitates the fusion of pertinent information between fused features, mitigating interference from redundant data. However, an excessively low downsampling dimension can amalgamate multiple key features, rendering them indistinguishable. The analyses in Fig. 10(b) and (c) show that the optimal downsampling dimension for both the GFFE and DFI modules in the Augsburg dataset is 32, whereas, in the Berlin dataset, the optimal downsampling dimensions for both the GFFE and DFI modules is 40. Due to the larger number of samples and complexity of the Berlin dataset, the downsampling dimension is higher in the Berlin dataset than in the Augsburg dataset, to extract more critical information for model learning.

4) Exponential Loss Function Parameters: The parameters α and β in the exponential label smoothing cross-entropy function significantly influence the penalty strength and smoothness of the loss function applied to the model. To assess their impact on the model's performance, we conduct experiments varying the values of α and β independently within the ranges of 1–10 and 0.1–0.9, respectively. The experimental findings, illustrated in Fig. 11(a) and (b), highlight distinct optimal values for α and β. Specifically, in the Augsburg and Berlin datasets, we observe that for the parameter α, the model reaches its best performance when the parameter value is set to 8. In contrast, for parameter β, the most efficient model training occurs at a parameter value of 0.6. While the exponential loss function is beneficial for enhancing the model's comprehension of intricate scenarios, excessive penalization may induce over-reliance on individual labels in the training data, thereby exacerbating overfitting issues. Introducing smoothing mechanisms to the exponential loss function not only mitigates the risk of overfitting but also introduces uncertainties in remote sensing image classification tasks. Specifically, label smoothing facilitates the acquisition of smoother decision boundaries by the model, thereby augmenting its robustness.



(a)

(b)

(c)

Fig. 12. OA (%) obtained by the weights of different loss function proportions on the Augsburg dataset. (a) Weight of the spatial–spectral fusion feature. (b) Weight of the global fusion feature.

5) Multiple Loss Function Weight Ratio: In the context of multibranch loss functions, each function is assigned a weight that reflects its relative importance and focus during model training. To explore the impact of these weights, we conduct experiments varying the values of a, b, and c within the range of 0.1–1. The experimental outcomes, depicted in Fig. 12(a)–(c), highlight the substantial influence of a, b, and c values on model training. Among these parameters, the value of a at 0.2 yields optimal training effects, while b and c demonstrate better training outcomes at a value of 0.9. The allocation of distinct weights to various loss functions can significantly influence the model's inclination toward learning features at different levels of abstraction. Illustrated by the chart, on the Augsburg dataset, the model exhibits a propensity for learning shallower features predominantly. Moreover, escalating the number of model layers impairs the pace at which the GFFE and SFFE modules extract crucial feature information, potentially impeding model efficiency. Consequently, augmenting the weights assigned to parameters "b" and "c" stands as an effective strategy to enhance the feature extraction velocity of GFFE and SFFE modules, thereby bolstering subsequent model training efficacy.

D. Performance Comparison

We conduct a comparative analysis of our model against contemporary state-of-the-art methodologies, including S^2ENet [39], MFT [20], Cross-HL [21], ExViT [22], fusion-HCT [23], and SSMAE [24].

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TABLE III
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS FOR THE AUGSBURG DATASET

Class	S^2ENet	MFT	CrossHL	ExViT	Fusion-HCT	SS-MAE*	SS-MAE	BPFT
Forest	98.10	97.37	98.59	90.04	97.32	96.82	96.48	**99.06**
Residential area	**99.08**	95.68	96.63	95.44	98.28	98.14	97.77	97.53
Industrial area	12.19	72.64	74.93	34.58	67.12	55.32	74.77	**87.23**
Low plants	91.78	96.79	97.84	90.68	93.33	95.94	95.87	**98.42**
Allotment	45.12	91.59	91.59	51.82	56.41	45.80	48.09	**95.98**
Commercial area	1.22	22.22	16.24	**28.63**	8.42	14.29	9.45	18.68
Water	24.09	60.92	58.00	17.65	49.77	46.21	43.52	**67.82**
OA	88.22	92.96	93.82	86.65	91.75	91.59	92.27	**95.34**
AA	53.08	76.74	76.26	58.40	67.24	64.64	66.56	**80.68**
Kappa	82.61	89.95	91.14	80.79	88.12	87.67	88.68	**93.32**

TABLE IV
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS FOR THE BERLIN DATASET

Class	S^2ENet	MFT	CrossHL	ExViT	Fusion-HCT	SS-MAE*	SS-MAE	BPFT
Forest	81.09	89.50	89.12	78.01	74.58	79.53	79.35	**89.77**
Residential area	73.05	82.42	88.31	74.05	72.58	70.99	76.04	**93.04**
Industrial area	62.61	**90.91**	72.85	39.48	48.92	63.06	68.51	79.39
Low plants	82.82	**92.13**	91.29	84.15	83.04	80.19	79.17	91.64
Soil	86.41	**95.29**	89.47	88.03	75.24	92.87	91.59	94.37
Allotment	54.61	**88.01**	57.98	70.00	76.34	64.60	56.86	85.24
Commercial area	2.56	**66.33**	42.81	38.18	29.66	26.37	19.75	54.90
Water	75.96	93.54	79.30	56.41	78.50	68.40	66.87	**96.29**
OA	71.08	84.78	83.46	72.63	71.18	71.15	73.43	**89.76**
AA	64.88	**87.27**	84.97	66.04	67.36	68.24	67.26	85.58
Kappa	58.02	77.04	75.43	60.48	58.66	59.00	60.91	**83.69**

1) **S^2ENet** introduces a spatial–spectral enhancement module designed to emphasize the complementary semantic content inherent in HSI and SAR data before fusion.
2) **MFT** utilizes the ViT model for multimodal remote sensing image classification, integrating both HSI and LiDAR data into tokens within the Transformer architecture for classification purposes.
3) **Cross-HL** introduces the cross HSI and LiDAR attention module to enhance understanding of land-use and land-cover (LULC) patterns.
4) **ExViT** introduces a new cross-modality attention module to promote information exchange across heterogeneous modalities, which exploits pixel-level spatial correlation in remote sensing scenes, and a token-based decision-level fusion module is used to attain further performance increase.
5) **Fusion-HCT** integrates a hierarchical CNN with a Transformer network to extract features from multisource data. Subsequently, it combines these extracted features using a cross-token attention fusion encoder.
6) **SSMAE** introduces a spatial–spectral masked autoencoder designed to leverage both spatial and spectral representations of HSI and LiDAR data. This is achieved through masking spatial patches and spectral channels, followed by a process of recovery and reconstruction.

1) Result on the Augsburg Dataset: Table III indicates that our model exhibits substantial enhancements compared to the ExViT model, with an 8.35% improvement in OA. Moreover, when compared to pretrained SSMAE, our model demonstrates a notable improvement of 2.73% in OA. Furthermore, our model's classification performance in the forest, industrial, low plants, allotment, and water regions has also seen enhancements, particularly in the water and industrial regions where significant improvements in classification performance are observed. Table III reveals a notable advancement in accuracy achieved by our model over the counterpart employing a Transformer backbone with self-attention, showcasing an improvement of approximately 3.1% even when juxtaposed against the utilization of pretrained SSMAE. The visualized classification results are shown in Fig. 9. This marked enhancement can be attributed to three pivotal factors. First, our model embraces a DFFI module, thereby furnishing a more comprehensive data landscape to the model while also facilitating the supplementation of pertinent information, thus optimizing the utilization of saliency features. Second, the integration of ESLoss, a novel loss function, fortifies the model's adeptness in navigating intricate scenarios, concurrently augmenting its capacity for generalization. Lastly, the adoption of the Fourier transform instead of the self-attention mechanism inherent in Transformers significantly reduces the parameter space, thereby streamlining the convergence process for the model.

2) Result On the Berlin Dataset: As depicted in Table IV, our model exhibits notable advancements in the Berlin dataset. Relative to MFT and CrossHL, our model's performance increases by 4.98% and 6.3%, respectively. Moreover, in comparison to S^2ENet, ExViT, Fusion HCT, SSMAE*, and

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TABLE V
INFLUENCE OF THE SFFE, GFFE, AND DFI MODULES ON THE BPFT MODEL

module			Augsburg			Berlin		
SFFE	GFFE	DFI	OA	AA	Kappa	OA	AA	Kappa
×	×	×	95.04%	79.97%	92.90%	84.82%	87.61%	77.49%
✓	×	×	94.58%	79.17%	92.22%	86.61%	**88.72%**	79.87%
×	✓	×	95.10%	80.34%	92.97%	87.02%	88.45%	80.44%
×	×	✓	95.26%	80.06%	93.20%	86.71%	87.80%	79.94%
✓	✓	×	95.02%	79.91%	92.86%	88.13%	87.30%	81.74%
✓	×	✓	95.01%	79.79%	92.84%	86.87%	88.31%	78.85%
×	✓	✓	95.28%	80.78%	93.22%	87.32%	87.87%	80.71%
✓	✓	✓	**95.34%**	**80.68%**	**93.32%**	**89.76%**	85.58%	**83.69%**

TABLE VI
COMPARISON OF CROSS-ENTROPY AND ESLOSS ON THE BPFT MODEL

Loss Function	Augsburg			Berlin		
	OA	AA	Kappa	OA	AA	Kappa
Cross-Entropy	95.12%	80.38%	93.01%	88.67%	**87.32%**	82.46%
Exponential Loss	95.05%	80.08%	92.90%	89.05%	86.36%	82.83%
Smoothing Loss	94.71%	77.56%	92.38%	83.97%	86.71%	76.26%
ESLoss	**95.34%**	**80.68%**	**93.32%**	**89.76%**	85.58%	**83.69%**

TABLE VII
EFFECTIVENESS OF THE DFFI MODULE

Method	Augsburg			Berlin		
	OA	AA	Kappa	OA	AA	Kappa
Single Branch	93.96%	79.06%	91.35%	84.25%	**86.59%**	76.63%
DFFI	**95.34%**	**80.68%**	**93.32%**	**89.76%**	85.58%	**83.69%**

TABLE VIII
EFFECTIVENESS OF THE MULTIBRANCH LOSS FUNCTION

Loss Function Branch		Augsburg			Berlin		
SFFE	GFFE	OA	AA	Kappa	OA	AA	Kappa
×	×	94.40%	79.68%	91.98%	86.23%	82.17%	78.41%
✓	×	94.89%	80.05%	92.67%	86.60%	79.27%	78.65%
×	✓	94.88%	80.23%	92.67%	88.67%	**88.01%**	82.50%
✓	✓	**95.34%**	**80.68%**	**93.32%**	**89.76%**	85.58%	**83.69%**

SSMAE models, our model showcases even more substantial improvements, with enhancements of 18.68%, 17.13%, 18.58%, 18.61%, and 16.33%, respectively. The classification map of Berlin dataset is shown in Fig. 8.

E. Discussion

1) Effectiveness of SFFE, GFFE, and DFI Modules: To evaluate the impact of the SFFE, DFI, and GFFE modules on the BPFT model, we conduct experiments using the Augsburg and Berlin datasets. The results, summarized in Table V, demonstrate the varying influence of these modules on model performance. Due to the complexity of the Berlin data compared to the Augsburg data, ablation experiments on different modules had a more pronounced effect on the results of the Berlin dataset. In the Berlin dataset, the OA was improved by 1.79%, 2.18%, and 1.89% compared to the baseline OA under retaining the SFFE module, the GFFE module, and the DFI module, respectively. In contrast, the OA values decreased by 2.44%, 2.89%, and 1.63% under ablating only the SFFE module, the GFFE module, and the DFI module, respectively. Notably, the SFFE and GFFE modules exerted a greater influence on model accuracy compared to the DFI module. This disparity may be attributed to the heightened significance of hierarchical feature extraction over feature interaction mechanisms in this experimental configuration.

2) Comparison of Loss Functions: To assess the effectiveness of our newly proposed loss function in model training, we perform comparative experiments on the Augsburg and Berlin datasets using traditional cross-entropy, exponential loss, and label smoothing cross-entropy as baselines. The experimental results, summarized in Table VI, illustrate the impact of these loss functions on model accuracy. On the Augsburg dataset, the exponential loss function and label smoothing cross-entropy lead to reductions in OA by 0.07% and 0.41%, respectively, compared to traditional cross-entropy. In contrast, our proposed exponential label smoothing loss function achieves an improvement of 0.22%. Similarly, on the Berlin dataset, the exponential loss function improves OA by 0.38%, while the label smoothing cross-entropy results in a significant accuracy drop of 4.7%. Our proposed exponential label smoothing loss function, however, yields a notable

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accuracy gain of 1.09%. While traditional cross-entropy yields relatively good performance, both the exponential loss and label smoothing cross-entropy show instability, leading to reduced accuracy in certain cases. In contrast, our proposed exponential label smoothing cross-entropy not only significantly improves model accuracy but also demonstrates greater robustness to variations in internal parameters.

3) Effectiveness of the DFFI Module: To validate the effectiveness of our proposed DFFI architecture, we compare it with the traditional single-branch approach to processing fusion features. The experimental results are shown in Table VII. Compared with the traditional single branch, the DFFI module improves the accuracy of the model by 1.38% on the Augsburg dataset and 5.51% on the Berlin dataset.

4) Effectiveness of the Multibranch Loss Function: To validate the effectiveness of the proposed multibranch loss function, we conduct an ablation study by removing the loss functions of the SFFE and GFFE branches from the original setup, retaining only the primary branch loss function. The results, summarized in Table VIII, demonstrate that the multibranch loss function improves accuracy by 0.94%, 1.00%, and 1.34% on the Augsburg dataset and by 3.53%, 3.41%, and 5.28% on the Berlin dataset, respectively. These findings indicate that the multibranch loss function facilitates the effective learning of both shallow and deep features, thereby enhancing the model's robustness.

V. Conclusion

In this article, we propose a novel model, termed BPFT, which demonstrates superior accuracy over existing approaches on two distinct datasets by effectively HSI and SAR data. The model architecture consists of two specialized branches designed to capture distinct, yet complementary fusion features. The first branch leverages an SFFE module, which focuses on enhancing spatial and spectral characteristics, while the second branch employs a GFFE module to extract global-level fusion features. These complementary modules work in tandem to improve feature extraction. To further refine the extraction process, we introduce learnable filters tailored to address frequency-domain features, enhancing the model's capacity to capture diverse patterns. Additionally, the DFI module facilitates the exchange of information between fused features, thus boosting the model's learning efficiency for key feature representations. Moreover, we propose an innovative exponential label smoothing cross-entropy loss function, designed to enhance the loss function's sensitivity to underrepresented category labels, particularly in complex scenes. This adjustment significantly improves the model's classification performance by better aligning the loss function with semantic correlations. Our experimental results confirm the effectiveness of the BPFT model in extracting comprehensive semantic features, leading to robust classification performance across various scenarios.

The presence of an imbalanced label distribution within the dataset impacts the learning capabilities of models, particularly affecting categories with fewer labeled instances, resulting in diminished learning performance. Consequently, our future research will prioritize addressing the challenges posed by multimodal classification tasks within datasets exhibiting uneven label distributions.

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